Dae Yon Hwang

Toronto, ON, Canada (Open to Relocation)
Email: ✉️ **Linkedin:** 🔗 **Personal Website:** 🔗

EDUCATION

University of Toronto	Ph.D. in Electrical & Computer Engineering, 4.0/4.0	Nov 2022
Texas A&M University	Master of Science in Electrical Engineering, 4.0/4.0	May 2016
Hanyang University	B.S. in Electronic Engineering, 3.56/4.0 ***(Cum Laude)***	Feb 2014

WORK EXPERIENCE

Layer 6 AI @TD - Senior Research Machine Learning Scientist *Trustworthy AI* May 2025 - Present

- **Black-Box Agent Optimization**
  - Developed a **Sequential Monte Carlo framework** for test-time policy optimization in black-box LLM agents, bridging reinforcement learning and Bayesian inference
  - Trained a **lightweight critic model** to guide black-box LLM policies, **reducing optimization cost by 20%** while **improving performance by 16%** over the GRPO baseline across agentic tasks. Published research findings in **ICML 2026**
- **Conversational RAG with Safety Guardrails**
  - Deployed a production RAG system balancing model capability with strict banking compliance requirements
  - Designed query classification, rewriting, and abstention mechanisms to enforce policy-aligned responses
  - Implemented a **three-layer safety framework** and led **automated agentic red-teaming** to mitigate adversarial inputs and jailbreak attempts
  - Launched a public customer-facing chatbot on TD Insurance, maintaining **zero jailbreak incidents** in production
  - Formulated multi-turn safety guardrails with **conformal prediction guarantees** under dynamic risk bounds, with findings submitted to **ICLR 2027**
- **Vision-Language Model Interpretability**
  - Built a synthetic pipeline for **counterfactual chart generation** to evaluate chart understanding in VLMs
  - Uncovered **memorization** and **shortcut learning behaviors** using both **heuristic benchmarking** and **mechanistic interpretability**. Published research findings in **EMNLP 2026**
- **Multi-Objectives Preference Optimization**
  - Formulated data topology based on **partially ordered sets** to model trade-offs in multi-objective preference alignment (e.g., helpfulness vs. harmlessness)
  - Evaluated across **DPO** and **listwise DPO**, achieving significant gains in alignment stability across up to **4 distinct objectives**. Under review at **ACL Rolling Review (ARR)** for upcoming ACL cycles

Amazon Science - Applied Scientist II *Artificial General Intelligence* Sep 2022 - Apr 2025

- **Multimodal Foundation Model Post-Training**
  - Generated **synthetic multi-step reasoning data** for post-training multimodal foundation models
  - Developed **LLM-as-a-Judge pipelines** for data filtering and quality control in post-training
  - Designed and scaled **fine-tuning** and **preference-optimization methods** (e.g., SFT, DPO, PPO, GRPO) across multi-GPU clusters using **TRL** and **FSDP**
  - Contributed to post-training **Amazon Nova foundation models**, improving performance across general and domain-specific tasks including legal and finance. Named contributor on the **Technical Report and Model Card**
- **Foundation-Model Recommendation System**
  - Developed and deployed a foundation-model recommendation system combining **instruction tuning**, **preference optimization**, and **RAG**. Validated the system through customer-facing experiments
  - Deployed the system on Prime Video, achieving a **20% improvement in positive-feedback rate** over the control model in online experiments

- **Information Retrieval for Alexa**
  - Developed **LLM-based synthetic-data generation** and **representation-learning methods** to improve zero-shot retrieval across diverse scenarios
  - Designed **dimensionality-reduction methods** that **accelerated retrieval efficiency by 5×** while maintaining performance. Published **3 papers** in top-tier venues, including **ACL** and **EMNLP**

University of Toronto - Research Assistant *Biometrics Security Lab* Sep 2018 - Sep 2022

- **Robust User Verification from Physiological Signals**
  - Developed **multimodal physiological-signal models** for robust user verification and adversarial resilience
  - Collected and released physiological signals from **170 participants**, creating a large public dataset for biometric research. Published **5 papers** in top-tier venues, including **IEEE TIFS** and **ICASSP**
- **Human Activity Recognition with Wearable Device**
  - Developed **hierarchical multimodal models** combining inertial and physiological sensors for wearable activity recognition. Published research findings in **ICASSP**

SELECTED PUBLICATIONS

D. Y. Hwang, et al., "Agentic Monte Carlo: Simulating Reinforcement Learning for Black-Box Agents" **ICML 2026**

Y. Jiang, **D. Y. Hwang**, et al., "Chartographer: Counterfactual Chart Generation for Evaluating Vision-Language Models" **EMNLP 2026**

D. Y. Hwang (*Contributor*), "The Amazon Nova Family of Models: Technical Report and Model Card" **Amazon Science 2024**

D. Y. Hwang, et al., "Link, Synthesize, Retrieve: Universal Document Linking for Zero-Shot Information Retrieval" **EMNLP 2024**

D. Y. Hwang, et al., "EmbedTextNet: Dimension Reduction with Weighted Reconstruction and Correlation Losses for Efficient Text Embedding" **ACL 2023**

D. Y. Hwang, et al., "GAN-LM: Generative Adversarial Network using Language Models for Downstream Applications" **INLG 2023**

D. Y. Hwang, et al., "PBGAN: Learning PPG Representations from GAN for Time-Stable and Unique Verification System" **IEEE TIFS 2021**

D. Y. Hwang, et al., "Evaluation of the Time Stability and Uniqueness in PPG based Biometric System" **IEEE TIFS 2020**

SELECTED RESEARCH TALKS

From Generative AI to Agentic AI: Building Trustworthy AI Kyungpook National University

Introduction to Mechanistic Interpretability TD Bank

Integrating GANs with Language Models Amazon

PROFESSIONAL SERVICE

Area Chair, Reviewer: EMNLP**,** ACL, ARR, NeurIPS, IEEE JBHI, IEEE TIFS

Program Committee: EMNLP 2023 Industry Track

SKILLS

Research: Foundation Models, LLM Post-Training, Reinforcement Learning, Agentic AI, Reasoning, Multimodal Learning, Trustworthy AI, Mechanistic Interpretability, Safety & Guardrails

Engineering: Python, PyTorch, C++, vLLM, Ray, Hugging Face (TRL, PEFT, Accelerate), FSDP, DDP, AWS, CUDA

Languages: English (Native/Fluent), Korean (Native)